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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 09/26/2024 |
| **PI Name:** | Mosier, Kathleen Ph.D. | | |
| **Project Title:** | Negotiating Crew Autonomy During Space Operations | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | (1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 94618-1542 | **Congressional District:** | 12 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity |
| **Start Date:** | 12/01/2023 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 09/30/2025 per A. Beitman/NASA JSC (Ed., 4/24/25).<br>NOTE: End date changed to 06/30/2025 per A. Beitman/NASA JSC (Ed., 2/14/25).<br>NOTE: End date changed to 02/28/2025 per L. Dang/NASA JSC (Ed., 8/5/24).<br>NOTE: Period of performance changed from initial award. Original start date: 11/17/2023. Original end date: 11/16/2024. New start and end dates changed per NSSC information (Ed., 2/4/2024). | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Fischer, Ute Ph.D. ( Georgia Institute of Technology )<br>Marquez, Jessica Ph.D. ( NASA Ames Research Center ) | | |
| **Grant/Contract No.:** | 80NSSC24K0442 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Space-ground collaboration during Artemis and other long missions will continue to be a requirement, given the complexity of space missions and unforeseen events threatening mission safety and success. It is important that crewmembers and mission support on Earth have a common understanding of crew autonomy and operational constraints, and employ a fluid approach to crew autonomy in areas such as the scheduling of tasks and activities. Crew self-scheduling is an integral component of crew autonomy and can serve as a proxy for other autonomy concerns and provide insights that can be applied to other components of autonomy, as it embodies characteristics that may cause friction between space crews and ground, undermine the accuracy of space-ground shared mental models, and negatively impact space-ground collaboration.<br>AIMS of the proposed work are to identify areas of contention in scheduling or where communication concerning scheduling may not be effective, and provide recommendations for how scheduling, as well as other aspects of autonomy could be negotiated and managed in a communication or decision support tool.<br><br>A review of autonomy issues in space operations, supplemented by selective data on crew autonomy and scheduling from the NASA Human Exploration Research Analog (HERA) and the NASA Scientific International Research In a Unique terrestrial Station (SIRIUS), will deliver: identification of problematic aspects of operational autonomy in space, with particular focus on Artemis; recommendations for how scheduling of tasks and activities can be negotiated, and best practices for managing shifting responsibilities; and definition of required features and parameters for a schedule negotiator technology function with feedback from selected subject matter experts.<br><br>SIGNIFICANCE of the proposed effort relates to its potential to facilitate collaboration and shared mental models among multi-team system members during Artemis and other long-term space operations. Successful scheduling negotiation is critical to mission success, and solutions will provide insight for other potentially disruptive autonomy elements. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | SIGNIFICANCE of the proposed effort relates to its potential to facilitate collaboration and shared mental models among space/ground multiteam system (MTS) members during Artemis and other long-term autonomous space crew operations. Successful negotiation of task and activity scheduling is critical to mission success and solutions will provide insight for other potentially disruptive autonomy elements. Recommendations for a decision support tool may be applicable for remote autonomous teams on earth. |
| **Task Progress:** | Space-ground collaboration will continue to be a requirement during Artemis and other long missions given the complexity of space missions and unforeseen events threatening mission safety and success. Nonetheless, long-distance space travel as well as establishing human outposts in space will inevitably require that a crew has more autonomy—i.e., self-sufficiency and self-directedness—than in current operations. The implementation of crew autonomy will represent a significant shift from current practice and introduce new challenges to space-ground collaboration. The roles and responsibilities of humans in space and humans on the ground will have to be redefined on the basis of a common understanding of crew autonomy and operational constraints. Moreover, future space exploration may necessitate a fluid approach to crew autonomy such that roles and responsibilities are negotiated throughout a mission rather than prescribed in advance.<br>A potential area of contention between autonomous space crews and ground support is the scheduling of tasks and activities. Crew self-scheduling will be an integral component of crew autonomy and can serve as a proxy for other autonomy concerns—it embodies characteristics that may undermine the accuracy of space-ground shared mental models, cause friction between space crews and ground, and negatively impact space-ground collaboration. AIMS of this effort, using crew self-scheduling as a proxy for issues and concerns associated with crew autonomy, are to: 1) identify areas of contention in scheduling or where communication concerning scheduling may not be effective; 2) provide recommendations for how scheduling in autonomous operations could be negotiated in the context of Artemis and other long-duration missions; and 3) propose human-centered design guidance for how this negotiation could be managed in a communication/decision support tool.<br><br>METHODS for this effort involve a multi-pronged approach including a review of autonomy issues in space operations, research on crew self-scheduling and literature relevant to the design of decision support for self-scheduling, interviews with autonomy researchers and NASA analog and operations personnel, and analyses of selective data on crew autonomy and scheduling from NASA analog missions.<br><br>During the past year, we identified relevant literature and created a database of categorized summaries of articles on autonomy, self-scheduling, multi-team systems, methodologies, and co-active/human-centered/ and ecological interface designs. These are helping us to define nuances and "trade spaces" of crew autonomy. We re-analyzed post-mission interviews with NASA analog crewmembers and ground support focusing on their understanding of crew autonomy and its implications for crew/ground collaboration. This provided insights into the meaning of autonomy and its motivating factors, crewmembers' scheduling preferences, and the crew/Mission Control relationship under autonomy.<br><br>We also conducted preliminary analyses of data in analog studies where participants were able to self-schedule some activities and characterized the particular types of activities participants chose for spontaneous self-scheduling. These activities included most often Operational activities such as habitat maintenance, followed by Outreach, Science, Exercise, Surveys, and Personal Time activities such as meals or pre-sleep.<br><br>We are currently conducting interviews with autonomy "experts," such as researchers, analog participants, flight crew, and ground support to contribute their insights on crew autonomy issues. Our questions focus on what autonomy means to the different stakeholders in space operations, what operational changes it would require, the challenges associated with crew self-scheduling, and how tasks and activities such as self-scheduling would be managed and negotiated during the course of an autonomous mission.<br><br>SIGNIFICANCE of this effort relates to its potential to facilitate collaboration and shared mental models among space-ground team members during Artemis and other long-term autonomous space crew operations. Successful negotiation of task and activity scheduling is critical to mission success and solutions will provide insight for other potentially disruptive autonomy elements. |
| **Bibliography Type:** | Description: (Last Updated: 02/11/2026) |
| **Conference Materials (Downloadable)** | Mosier K, Fischer U, Marquez J. "Negotiating crew autonomy during space operations." ASPIRE-The 69th Human Factors and Ergonomics Society Annual Meeting, Phoenix, Arizona, September 9-13, 2024. , Sep-2024 HFES2024 Poster - Negotiating Autonomy.pdf |
| **Significant Media Coverage** | Mosier K. "The future of space: Kathleen Mosier of Human Factors and Ergonomics Society on how space travel and the space industry can improve our lives here on Earth." Authority Magazine, August 1, 2024. https://medium.com/authority-magazine/the-future-of-space-kathleen-mosier-of-human-factors-and-ergonomics-society-on-how-space-travel-73815ed12aba , Aug-2024 |